

Functions

Definition

Given two sets A and B where $A, B \subseteq \mathbb{R}$,

A rule that assigns to each element to A to a unique element in B is called a function from A to B

A is the domain, B is the codomain, and for a function f , we write:

$$f: A \rightarrow B$$

or f maps A to B

$A(f)$ = domain and $B(f)$ = codomain

If $x \in D$ and $y \in R$ are related by the function f , we write $y = f(x)$

Here, $R = f(D)$ known as the range or image

$$R \equiv R(f)$$

Codomain = Possible outputs

Range = Actual outputs

Maximal domain = largest subset of $\mathbb{R}$ for which the formula can be evaluated to give a finite value

function cannot map to more than one output from single input

Properties of functions

Definition: Suppose f is defined on a subset of $\mathbb{R}$ which is symmetric about 0 ($x \in D(f) \Rightarrow -x \in D(f)$)

If $f(x) = f(-x)$ for all $x \in D(f)$, f is an **even** function

$f(x) = -f(-x) \Rightarrow$ odd function

$$O(f)(x) = \frac{1}{2} [f(x) - f(-x)]$$

$$E(f)(x) = \frac{1}{2} [f(x) + f(-x)]$$